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Enantioselective Synthesis of Multifunctionalized 4*H*-Pyrans via Formal [4 + 2] Annulation Process by Bifunctional Phosphonium Salt Catalysis

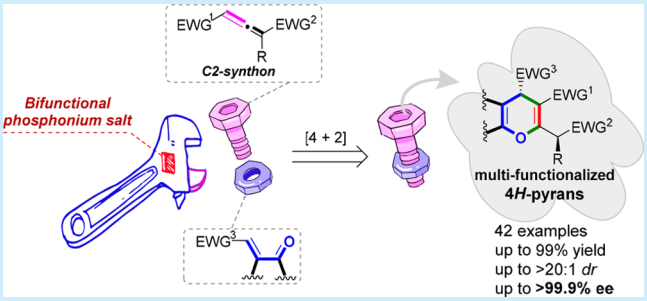
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Supporting Information

ABSTRACT: A highly enantioselective formal [4 + 2] annulation involving electron-deficient allenes as C2-synthons has been developed under bifunctional phosphonium salt catalysis. With this catalytic protocol, a wide range of synthetically interesting and highly functionalized chiral 4*H*-pyran derivatives were readily prepared in good yields (up to 99%) with outstanding diastereo- and enantioselectivities (up to >20:1 *dr*, up to >99.9% ee). The utility and the practicality of this method were demonstrated by the gram-scale reaction and facile elaboration. Of note, this is the first example of an asymmetric annulation reaction involving allenic reactants under a phase-transfer catalysis system.



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In the past decades, asymmetric PTC (phase transfer catalysis) has been known as a kind of rather powerful and efficient method for enantioselective preparation of structurally diverse and synthetically useful molecular architectures, and a number of quaternary ammonium salts have been commonly applied in PTC.[1] In contrast, asymmetric reactions catalyzed by chiral phosphonium salts are rare and still underdeveloped.[2] The pioneering works were recently conducted by the research groups of Ooi[3] and Maruoka.[4] Lately, Ma and co-workers disclosed a binol-derived P-spiro-phosphonium salt and applied it to an asymmetric amination reaction.[5] More recently, Zhao[6] and Lu[7] developed a few phosphonium salt catalysts derived from natural amino acids and further preliminarily realized their utilization in asymmetric conjugate addition and substitution reactions. Our group recently achieved enantioselective annulation reactions by designing and making use of novel dipeptide-based bifunctional phosphonium salt catalysts.[8] It must be noted that such type of bifunctional phosphonium salts is highly tunable, but their application in asymmetric organic synthesis is still rare.

On the other hand, electron-deficient allenes containing highly activated C=C bonds have been widely employed in asymmetric organic synthesis, especially in a wide range of asymmetric cyclization processes by a phosphine[9,10] or amine[11−13] catalysis system. However, their use in a phase transfer catalysis (PTC) system has been rather limited.[14] To our best knowledge, there are just two examples achieved by chiral ammonium salt catalysis (Figure 1a). In 2008, Jørgensen et al. utilized cyclic β-ketoesters and glycine imine derivatives as pronucleophiles in a conjugated addition with electron-deficient allenes by ammonium salt catalysis.[14a] In 2013, Maruoka and co-workers disclosed an ammonium salt-catalyzed asymmetric alkylation and alleno-Mannich-type reaction of 1-alkylallene-1,3-dicarboxylates.[14b] Consequently, developing other types of reactions involving allenic esters by PTC is highly desirable. In continuation of our major interest in bifunctional organocatalysis involving phosphonium salt catalysts, we questioned whether activated allenes could be efficiently used as C2/C3 synthons for an annulation process, in combination with α,β-unsaturated ketones as C4 synthons to produce structurally unique and biologically significant cyclic compounds (Scheme 1).[15] In this report, we described an asymmetric [4 + 2] annulation reaction involving allenes as C2-synthons for the synthesis of highly functionalized 4*H*-pyrans under a bifunctional phosphonium salt catalysis system (Figure 1b).

Initially, we began our studies via investigating the annulation reactions between 3-alkenyl-oxindole **1a** and allenic ester **2a** in the presence of a number of amino acid derived bifunctional phosphonium salt catalysts (Table 1). L-Valine-

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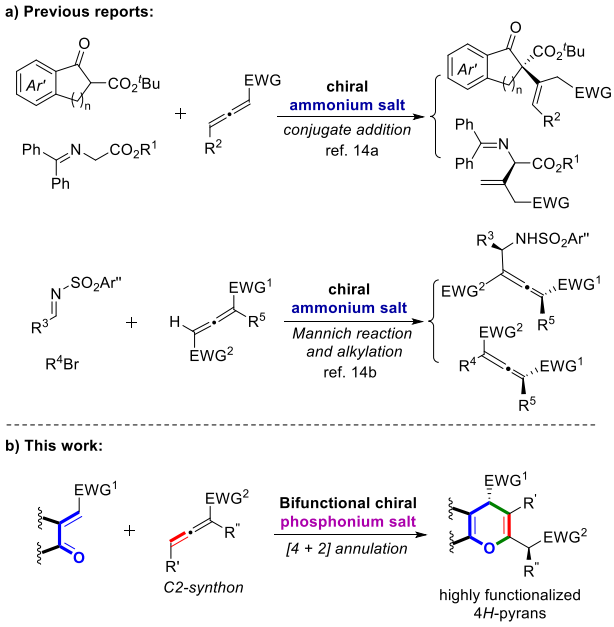


Figure 1. Asymmetric processes involving electron-deficient allenes under PTC system.

Scheme 1. Representative Bioactive 4*H*-Pyran-Containing Compounds

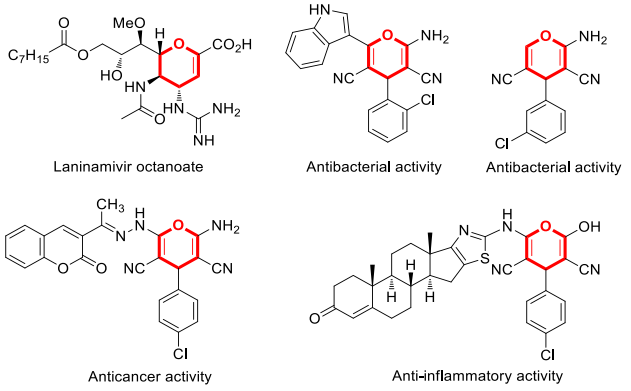
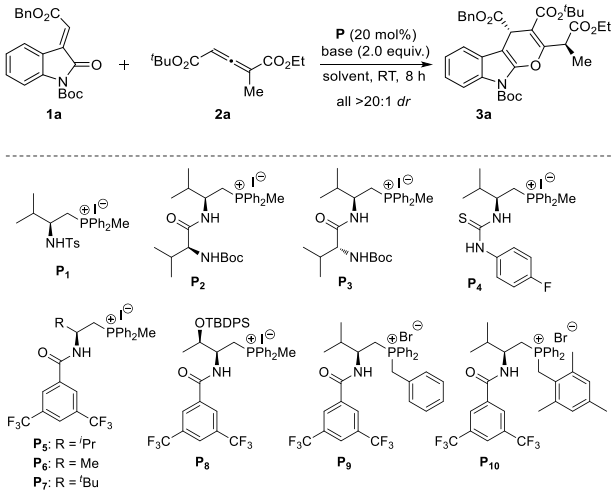


Table 1. Reaction Optimization[a]



entry	P	solvent	base	yield (%)[b]	ee (%)[c]
1	$\mathbf{P_1}$	toluene	Cs_2CO_3	76	15
2	$\mathbf{P_2}$	toluene	Cs_2CO_3	91	5
3	$\mathbf{P_3}$	toluene	Cs_2CO_3	90	3
4	$\mathbf{P_4}$	toluene	Cs_2CO_3	92	7
5	$\mathbf{P_5}$	toluene	Cs_2CO_3	97	50
6	$\mathbf{P_6}$	toluene	Cs_2CO_3	93	25
7	$\mathbf{P_7}$	toluene	Cs_2CO_3	96	14
8	$\mathbf{P_8}$	toluene	Cs_2CO_3	95	45
9	$\mathbf{P_9}$	toluene	Cs_2CO_3	98	73
10	$\mathbf{P_{10}}$	toluene	Cs_2CO_3	98	81
11	$\mathbf{P_{10}}$	toluene	K_2CO_3	89	83
12	$\mathbf{P_{10}}$	toluene	K_3PO_4	98	91
13	$\mathbf{P_{10}}$	xylene	K_3PO_4	98	93
14[d]	$\mathbf{P_{10}}$	xylene	K_3PO_4	98	93
15[d]	$\mathbf{P_{10}}$	xylene	none	n.r.	–

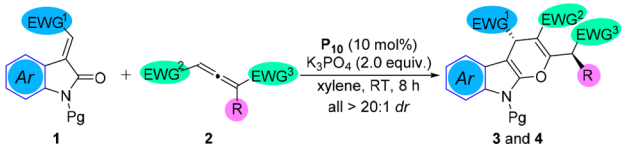
[a]Reactions were performed with **1a** (0.1 mmol), **2a** (0.12 mmol), catalyst (0.02 mmol), base (0.2 mmol) in solvent (2.0 mL) at room temperature for 8 h. All *dr* values were determined by ^{1}H NMR of crude product. [b]Isolated yields. [c]Determined by HPLC on a chiral stationary phase. [d]The catalyst loading was 10 mol %. TBDPS = *tert*-butyldiphenylsilyl, Ts = 4-toluenesulfonyl. n.r. = no reaction.

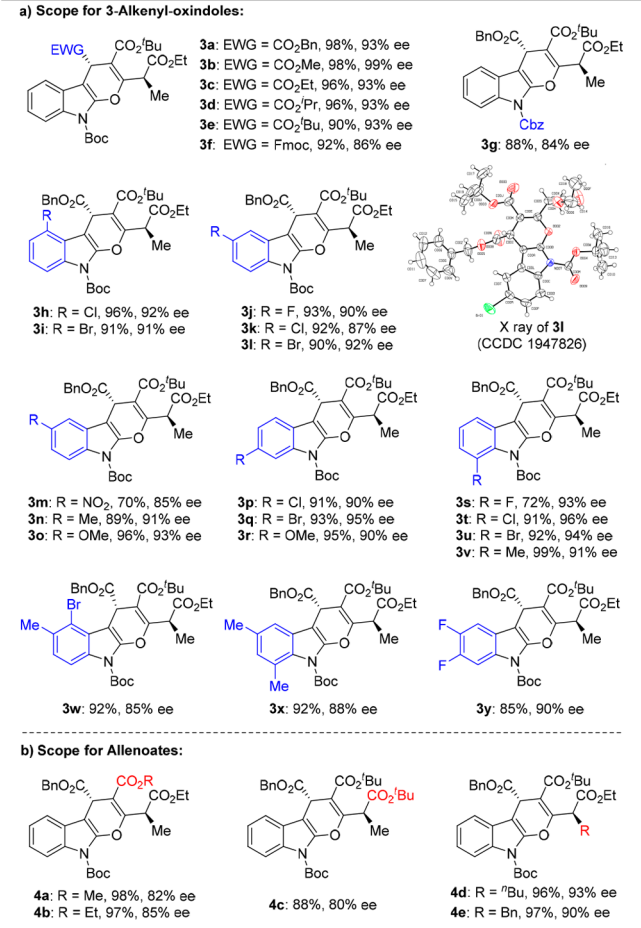
derived phosphonium iodide catalysts bearing a sulfonamide, dipeptide, thiourea, or amide unit were found to be effective for promoting this transformation to produce 4*H*-pyran **3a** in moderate to good isolated yields with excellent diastereoselectivities but with low enantioselectivities (entries 1−5). Changing the backbone to alanine, *tert*-leucine, or threonine did not improve the enantioselectivities (entries 6−8). While the L-valine-derived amide-based phosphonium bromides were performed, the ee value of this annulation reaction increased dramatically (entries 9 and 10), and the catalyst $\mathbf{P_{10}}$ tuned out to be the best (entry 10). Later, quick screenings of base (Table S2) and solvent (Table S3) indicated that K_3PO_4 was the optimal choice of base and the xylene was the suitable solvent (entries 12 and 13).[16] Indeed, the reaction also worked well with 10 mol % catalyst, affording the corresponding product in 98% yield and 93% ee (entry 14). Practically, there is no reaction when the base is absent (entry 15).

With the optimal reaction conditions in hand (Table 1, entry 14), we started to focus our attention on the scope of this [4 + 2] annulation between 3-alkenyl-oxindoles **1** and allenic ester **2** (Table 2). The ester substituents and *N*-protected groups of 3-alkenyl-oxindoles were first investigated. Pleasantly, various ester groups (EWG^1) on the 3-alkenyl-oxindoles were well tolerated, furnishing the corresponding products (**3a−f**) in excellent yields (90−98%) with good to excellent stereoselectivities (all >20:1 *dr*, 86−99% ee). *N*-Benzyloxycarbonyl 3-alkenyl-oxindole was also compatible to deliver the expected product **3g** in high yield (88%) with high stereoselectivities (>20:1 *dr* and 84% ee). Subsequently, various 3-alkenyl-oxindoles bearing electron-withdrawing, -donating, and -neutral groups on the phenyl ring were found to be compatible under the standard reaction conditions, and the products **3h−y** were produced in high yields (70−99%) with excellent diastereo- and enantioselectivities (all >20:1 *dr*, 85−96% ee). Of note, when the benzene rings bore electron-withdrawing substituents, the yields of annulation products were slightly lowered (**3m**, **3s**, and **3y**). The catalytic reactions also worked well when different allenic esters were employed, giving the expected products **4a−e** in high yields (88−98%) with excellent stereoselectivities (all >20:1 *dr* and 80−93% ee). The absolute configurations of these annulation products were

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Table 2. Substrate Scope[a]



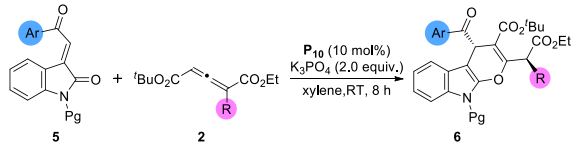


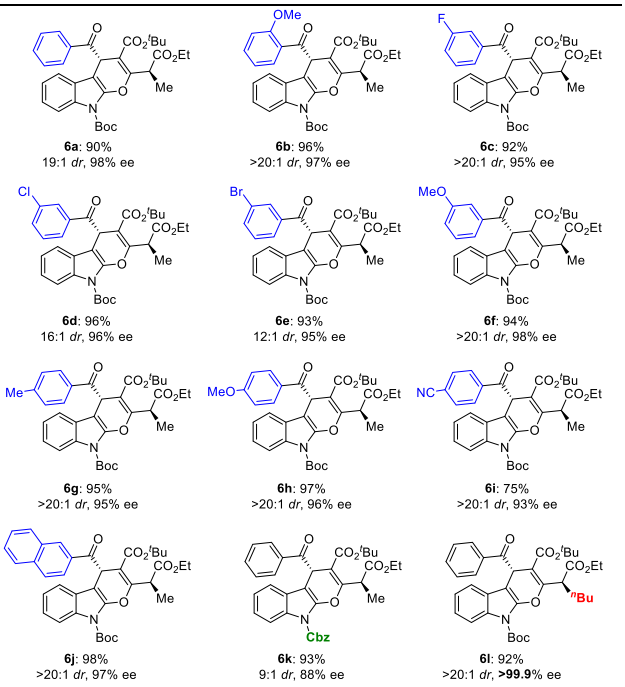
[a]Reactions were performed with **1** (0.2 mmol), **2** (0.24 mmol), P_{10} (0.02 mmol), and K_3PO_4 (0.4 mmol) in xylene (4.0 mL) at room temperature for 8 h. All dr values were determined by HPLC analysis of crude product.. Isolated yields. The ee values were determined by HPLC on a chiral stationary phase.

determined by X-ray crystal structural diffraction analysis of the product **3l** (CCDC 1947826).

Subsequently, we also investigated the [4 + 2] annulation between ketone-substituted 3-alkenyl-oxindoles and electron-deficient allenes (Table 3). Generally, a wide range of ketone-activated 3-alkenyl-oxindoles (**5**) with different substituents in the *para*-, *ortho*-, or *meta*-position on the phenyl ring of ketone-moiety proceeded rather well to deliver the desirable products **6a**−**j** in high yields (75−98%) with excellent stereoselectivities (up to >20:1 *dr* and 93−98% ee). When the *N*-protection group was changed to Cbz, the reaction also proceeded smoothly (**6k**). Moreover, *n*-butyl-substituted allenoate also proved to be an excellent substrate and afforded the product **6l** with an exceptionally high ee value of greater than 99.9% with 92% yield. Moreover, some gram-scale reactions were conducted to prepare the desirable products such as **3b** and **3l** with comparable yields as well as excellent diastereoselectivities and enantioselectivities (Figure S2a); further, the product **3l** was readily transformed into different optically enriched compounds **7** and **8** in high yields (Figure S2b).[16]

Table 3. Asymmetric [4 + 2] Annulation between Compounds 5 and Allenoates 2[a]

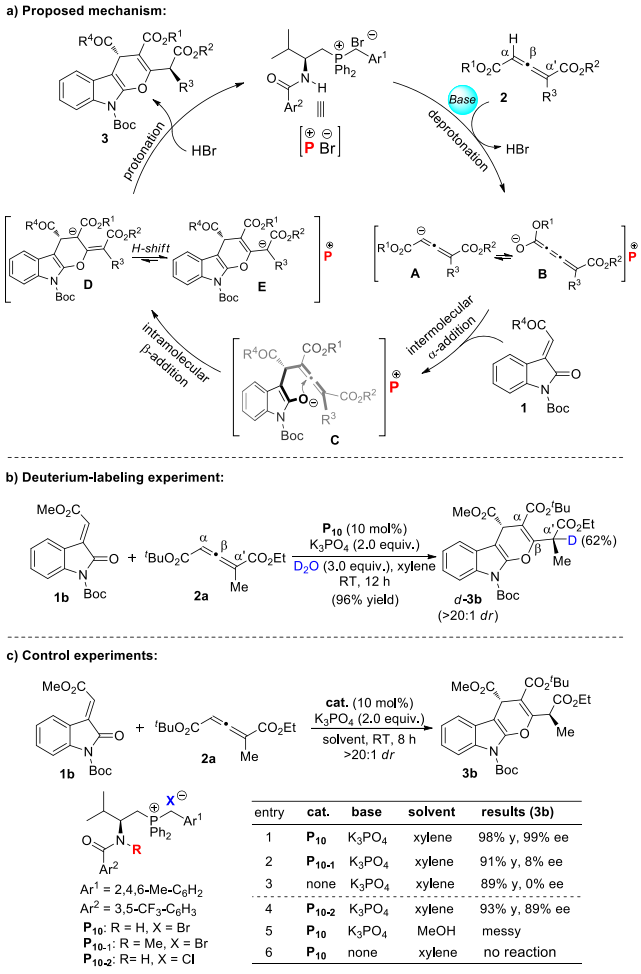




[a]Reactions were performed with **5** (0.2 mmol), **2** (0.24 mmol), P_{10} (0.02 mmol), and K_3PO_4 (0.4 mmol) in xylene (4.0 mL) at room temperature for 8 h. All dr values were determined by HPLC analysis of crude product. Isolated yields. The ee values were determined by HPLC on a chiral stationary phase.

Based on the obtained results and our previous explorations,[8] a plausible mechanism was presented in Figure 2a. We propose that the base-mediated deprotonation process initially occurs to generate the anionic intermediate **A**/**B**. Then, α-addition of this intermediate **A**/**B** to 3-alkenyl-oxindoles (**1**) leads to the key intermediate anionic **C** which may be stabilized by a phosphonium cation via ion-pair interaction. Intramolecular β-addition of **C** yields the intermediate **D** which can isomerize to form main intermediate **E**. Last, the anionic intermediate **E** abstracts a proton from the system to generate the desired product **3** and simultaneously release the phosphonium salt catalyst, which will enter the reaction cycle again. Accordingly, the deuterium-labeling experiment was performed to gain some preliminary insight into this catalytic cycle. When 3.0 equiv of D_2O were added into the standard reaction system, deuterium incorporation into the α'-position of product (*d*-**3b**) was observed, which suggests the beneficial existence of intermediate **E** in the catalytic cycle. To gain some insights into the origin of the stereochemical outcome, we further prepared the methylated catalyst $P_{10\text{-}1}$ and tested its catalytic ability for the model reaction. It was found that either the enantioselectivity decreased dramatically or racemic product was delivered in the absence of a chiral catalyst

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entry	cat.	base	solvent	results (3b)
1	P_{10}	K_3PO_4	xylene	98% y, 99% ee
2	$P_{10\text{-}1}$	K_3PO_4	xylene	91% y, 8% ee
3	none	K_3PO_4	xylene	89% y, 0% ee
4	$P_{10\text{-}2}$	K_3PO_4	xylene	93% y, 89% ee
5	P_{10}	K_3PO_4	MeOH	messy
6	P_{10}	none	xylene	no reaction

Figure 2. Mechanistic insights and proposed reaction mechanism.

(Figure 2c, entries 1−3). Moreover, the phosphonium chloride catalyst ($P_{10\text{-}2}$) still worked well for affording desired **3b** (Figure 2c, entry 4). When the reaction was conducted in methanol, the result was messy. Of note, the reaction could not occur when no base was added (Figure 2c, entry 6). These experimental results indicated that the weak-bond interactions, such as those related to ion pairs and H-bonding, were critical for the excellent stereoselectivities under such a bifunctional phosphonium system (Figure S3).[16]

In summary, an efficient alternative for the enantioselective synthesis of highly functionalized 4*H*-pyrans via bifunctional phosphonium salt catalyzed [4 + 2] annulation of electron-deficient allenes with 3-alkenyl-oxindoles has been disclosed. With this catalytic asymmetric protocol, a broad range of highly substituted 4*H*-pyran derivatives were prepared in high yields with excellent diastereo- and enatioselectivities under mild reaction conditions. These results indicated that this kind of structurally tunable bifunctional phosphonium salt catalysis was a practicable strategy to construct structurally complicated chiral compounds, and further investigation is ongoing in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.9b04079.

Experimental procedures and compound characterization data (PDF)

Accession Codes

CCDC 1947826 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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